

NAIROBI SCHOOL 2025 PREDICTION



KCSE ANTICIPATION EXAM

231/3

BIOLOGY

PAPER 3 (PRACTICAL)

TIME: 1¾ HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

(Kenya Certificate of Secondary Education)

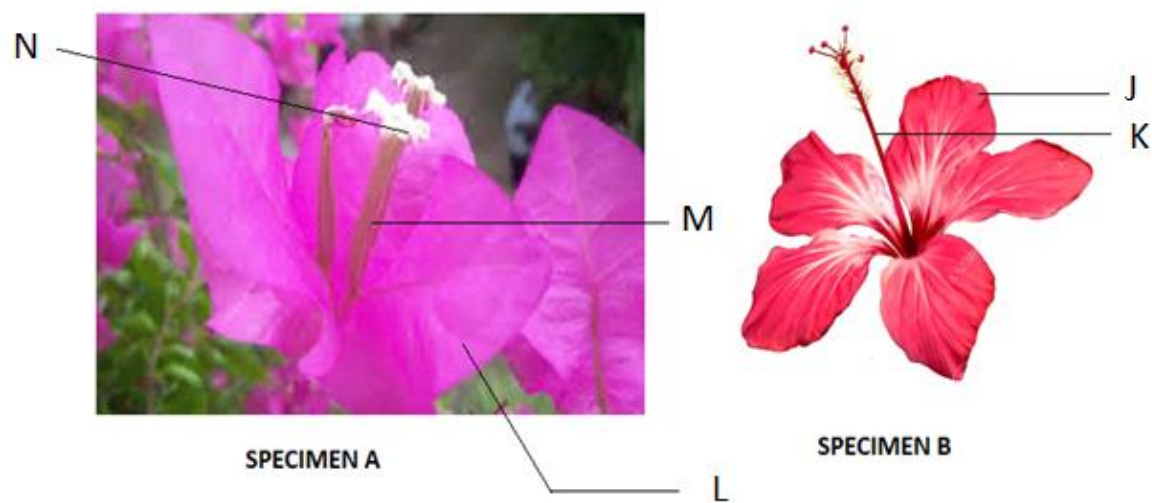
Instructions to Candidates

- Write your name, class, admission number, school, date and signature in spaces provided above.
- Answer **all** questions in the spaces provided below each question.
- You are required to take the first 15 minutes of the 1¾ hours allowed for this paper reading through the whole paper before commencing your work.
- Additional pages must not be inserted.
- Candidates should answer all the questions in English.

FOR EXAMINER'S USE ONLY

QUESTION	MAXIMUM SCORE	CANDIDATES SCORE
1	13	
2	14	
3	15	
Total Score	40	

1. You are provided with a photograph of two flowers, **A** and **B**. Observe the photograph carefully to answer the questions that follow.



- (a) Identify the parts labelled:
- K..... (1 mark)
- L..... (1 mark)
- M..... (1 mark)
- (b) State **two** observable differences between flower **A** and **B**. (2 marks)
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- (c) i) Name the agent of pollination in flower **A**. (1 mark)
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- (ii) How is it suited to the agent named in c(i) above? (1 mark)
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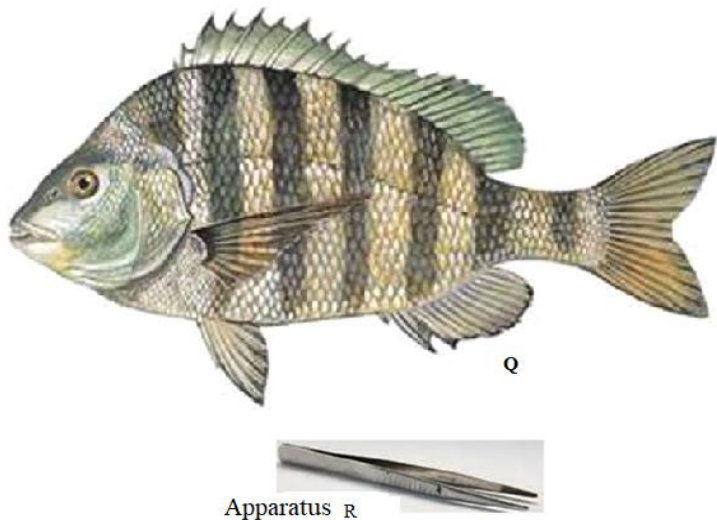
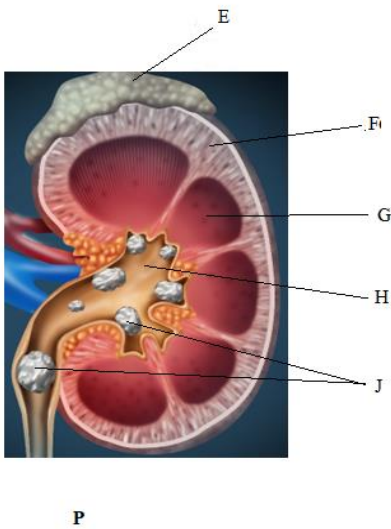
(d) You are provided with specimen **S** and **T**. Use them to answer the questions that follow.

i) Draw a plan diagram of the cut surface of specimen **S** and label it. (4 marks)

ii) With reason, identify the type of fruit **T**:

Identity	(1 mark)
.....	
Reason	(1 mark)
.....	
.....	

2. The photograph below shows longitudinal section through the kidney of animal **P** and fresh water animal **Q**.



a) Name the parts labelled **F**, **G** and **H**. (3 marks)

F.....

G.....

H.....

b) i) Name the nitrogenous waste you would expect organism **P** and **Q** to excrete. **(2 marks)**

P.....

Q

ii) Explain your answer for organism **Q** in **(b) (i)** above. **(2 marks)**

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c) The section of the kidney of organism **P** shows that the organism suffers from a common kidney disease.

i) Name the disease. **(1 mark)**

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ii) State two control measures for the disease. **(2 marks)**

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d) Explain the role of the part labelled **E** in coordination in animal. **(2 marks)**

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e) If the actual size of apparatus **R** is 50 mm, calculate the actual length of organism **Q**. (Show your working). **(2 marks)**

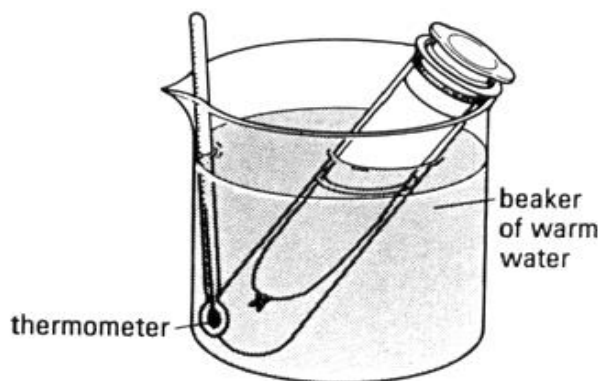
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3. You are provided with substance **L**, solution **M**, distilled water, one boiling tube, three test tubes, labels, one visking tubing, pieces of thread, Iodine solution and Benedict's solution.

i) Label the boiling tube **W**. Measure 20 ml of distilled water and put into the boiling tube.

Tie one end of one of the visking tubing with a piece of thread provided. With the help of a syringe, place 3 cm³ of substance **L** into the visking tubing then add three drops of Iodine solution. Record the observations in the table below. Using another syringe, add 3 cm³ of solution **M**. Tie the other end of the visking tubing tightly. Ensure there is no leakage on both ends of the visking tubing. Wash the outside of the visking tubing with water. Place the visking tubing upright in boiling tube labelled **W** with the 20 ml of distilled water.



Set up **K**: visking tubing in water in boiling tube

Immerse the boiling tube in hot water in a beaker maintained at 35 °C to 40 °C as shown above. Allow the set up to stand in hot water bath for 10 minutes. **Preserve the substance in the boiling tube (outside the visking)**. Record the observations in the table below. (2 marks)

Boiling tube	Observation	
	After adding Iodine solution	After 10 minutes in the water bath
W		

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- ii) Using the 2 ml of the substance outside the visking tubing test for the food substance using Benedict's solution provided. Record the observations in the table below. **(4 marks)**

Food substance	Procedure	Observations	Conclusion

a. Suggest the identity of:

- i) substance **L**. **(1 mark)**

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- ii) substance **M** in human beings. **(1 mark)**

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b. Name regions in human beings where a similar reaction occurs. **(2 marks)**

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c. Account for the observations made in boiling tube **W** in procedure (i) and (ii) above. **(3 marks)**

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